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1  //A non-recursive program for binary search:
2  # include <stdio.h>
3
4  //function prototype
5  void binarySearchIteration(int[],int,int,int,int);
6
7  int number[5]; //global declaration of array
8
9  //Function definition
10 void binarySearchIteration(int number[], int size, int low, int high, int data)
11 {
12     int mid,flag=0;
13     while(low <= high)
14     {
15         mid = (low + high)/2;
16         if(number[mid] == data)
17         {
18             flag = 1;
19             break;
20         }
21         else
22         {
23             if(data < number[mid])
24                 high = mid - 1;
25             else
26                 low = mid + 1;
27         }
28     }
29     if(flag == 1)
30         printf("\n %d found at location: %d", data, mid + 1);
31     else
32         printf("\n %d Not Found",data);
33 }
34
35 void main()
36 {
37     int size, data, i, low, high;
38     printf("\n Enter the number of elements: ");
39     scanf("%d", &size);
40     printf("\n Enter the elements in ascending order: ");
41     for(i = 0; i < size; i++)
42         scanf("%d", &number[i]);
43     printf("\n Enter the element to be searched: ");
44     scanf("%d", &data);
45     low = 0; high = size-1;
46     //function calling
47     binarySearchIteration(number,size,low,high,data);
48 }

```